Airbnb Marketing Analytics Final Project

Presenters: Clark Hayashi, Blake Dorn, Keshav Dhawan, Anthony Merlino, Ben Sakamoto

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Business Context & Problem

Our Midterm Research Question Was….

Where and when can post-grad travelers find the best value in Tokyo Airbnb stays?

Optimal grouping: Ota Ku (Near Airport), February, Entire Home

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Research Question

What separates a "cheap" Tokyo Airbnb from a truly high-value stay for post-grad travelers?

Role: Guest (Post-grad travelers evaluating Tokyo Airbnb value)

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Business Context & Problem

What will a high-value Tokyo stay look like when we combine pricing data with real guest feedback?

→Guest Review Variables

→Factor Analysis

→Cluster Analysis

Structured listing data: PCA, clustering, classification

Review text data: word count, sentiment, topic modeling

Data Overview & Preparation

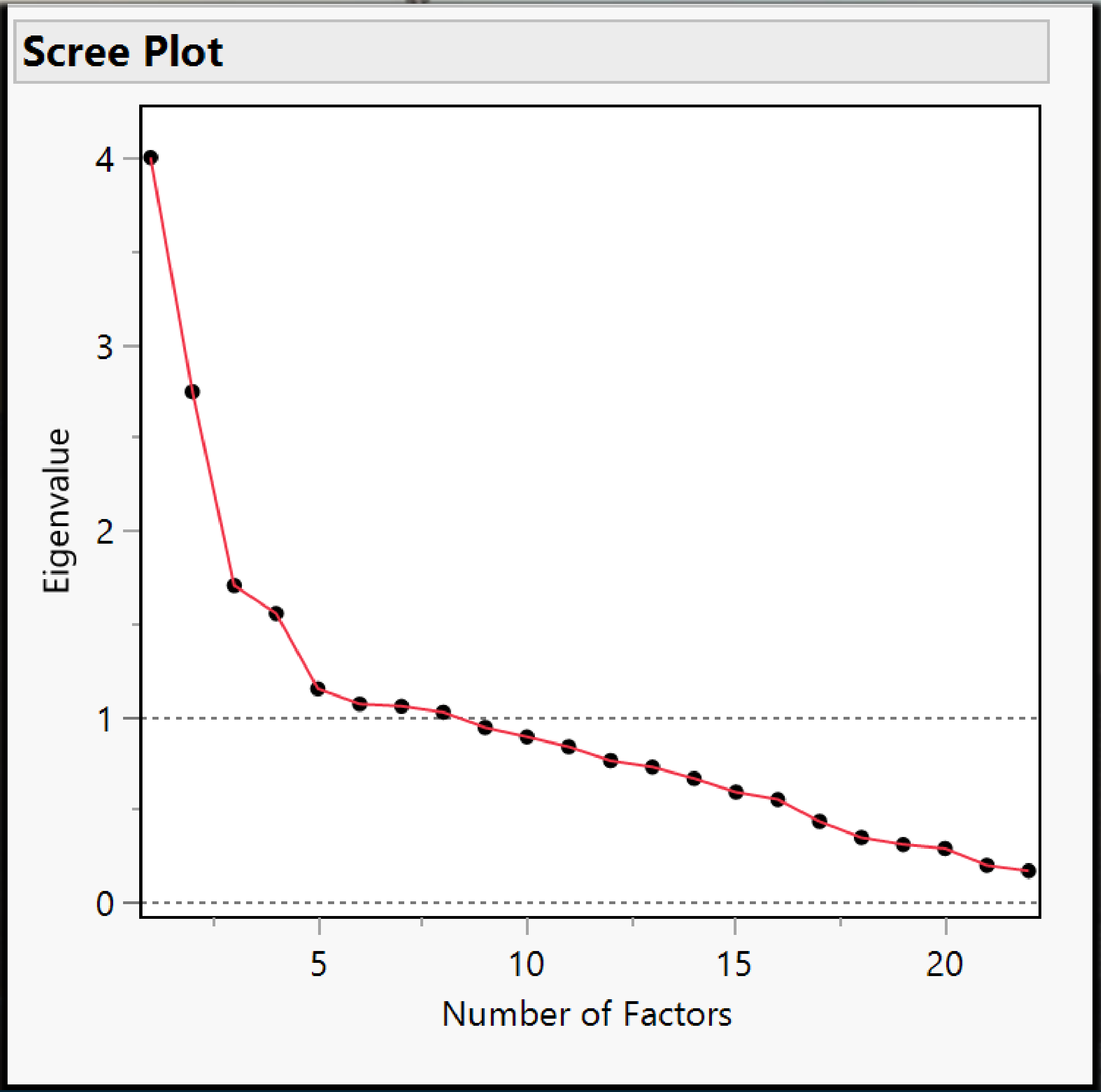
- Used the same cleaned data set from the midterm
  - Tokyo listings, around 25k modeled listing
- Added the review data analysis
- Join key: listing_id

Used resulting data in our final recommendation considerations

PCA Methodology

➢ What underlying listing dimensions explain differences across Tokyo Airbnbs?

- Principal Component Analysis (PCA) was used to reduce 21 Airbnb variables into a smaller set of interpretable dimensions.
- Variables included review scores, property characteristics, host attributes, pricing, and availability measures.
- Components with eigenvalues greater than 1 were retained
- Compiling the following explained ~51% of total variation



Factor Extraction

- Five interpretable factors were extracted with the use of varimax rotation
- These factors explained ~50.7% of variation in the characteristics of listings
- Two greatest factors are shown to be; Guest Satisfaction & Property Size

Variance Explained by Each Factor

Factor	Variance	Percent	Cum Percent
Factor 1	3.7275	16.943	16.943
Factor 2	2.8323	12.874	29.817
Factor 3	1.7838	8.108	37.926
Factor 4	1.4601	6.637	44.562
Factor 5	1.3554	6.161	50.723

Factor Labels & Regression

Factor	Interpretation
Factor 1	Guest Satisfaction(Ratings, Value, Cleanliness, Communication, Location
Factor 2	Property Size(Beds, Bedrooms, Bathrooms, Accommodates)
Factor 3	Listing Popularity(Reviews and Booking Activity)
Factor 4	Host Responsiveness(Response rate, Acceptance Rate, Instant Booking
Factor 5	Host Experience(Host Tenure, and Experience Listing)

Factor Labels & Regression

- Tokyo Airbnb Market
  - Guest Satisfaction (16.9%)
    - Listing Popularity(8.1%)
    - Host Responsiveness(6.6%)
  - Property Size (12.9%)
    - Host Experience(6.2%)

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Cluster Development

Cluster Means

Column Summary

Column	RSquare
host_tenure_years	0.1046
superhost_dummy	0.5201
host_response_rate_num	0.6248
host_acceptance_rate_num	0.6202
calculated_host_listings_count	0.6293
accommodates	0.5519
bathrooms_num	0.4501
bedrooms_clean	0.5193
beds_clean	0.4755
amenities_count	0.1595
price_per_guest	0.8737
minimum_nights	0.7664
maximum_nights	0.6247
instant_bookable_dummy	0.5875
availability_365	0.0220
number_of_reviews	0.3905
reviews_per_month_clean	0.4404
review_scores_rating	0.5547
review_scores_location	0.3213
review_scores_value	0.5000
review_scores_cleanliness	0.4510
review_scores_communication	0.4364

Portion of total variation in each column absorbed by clustering

Key Findings

- Hierarchical clustering identified distinct groups of Airbnb listings in Tokyo.
- Price per Guest and Minimum Nights were the strongest variables driving cluster separation.
- Host Response Rate, Host Acceptance Rate, and Number of Host Listings were also key differentiators.
- Property characteristics and review scores contributed to segmentation patterns.
- The results demonstrate that pricing strategy, host quality, and listing features are the primary factors distinguishing Airbnb listings in Tokyo

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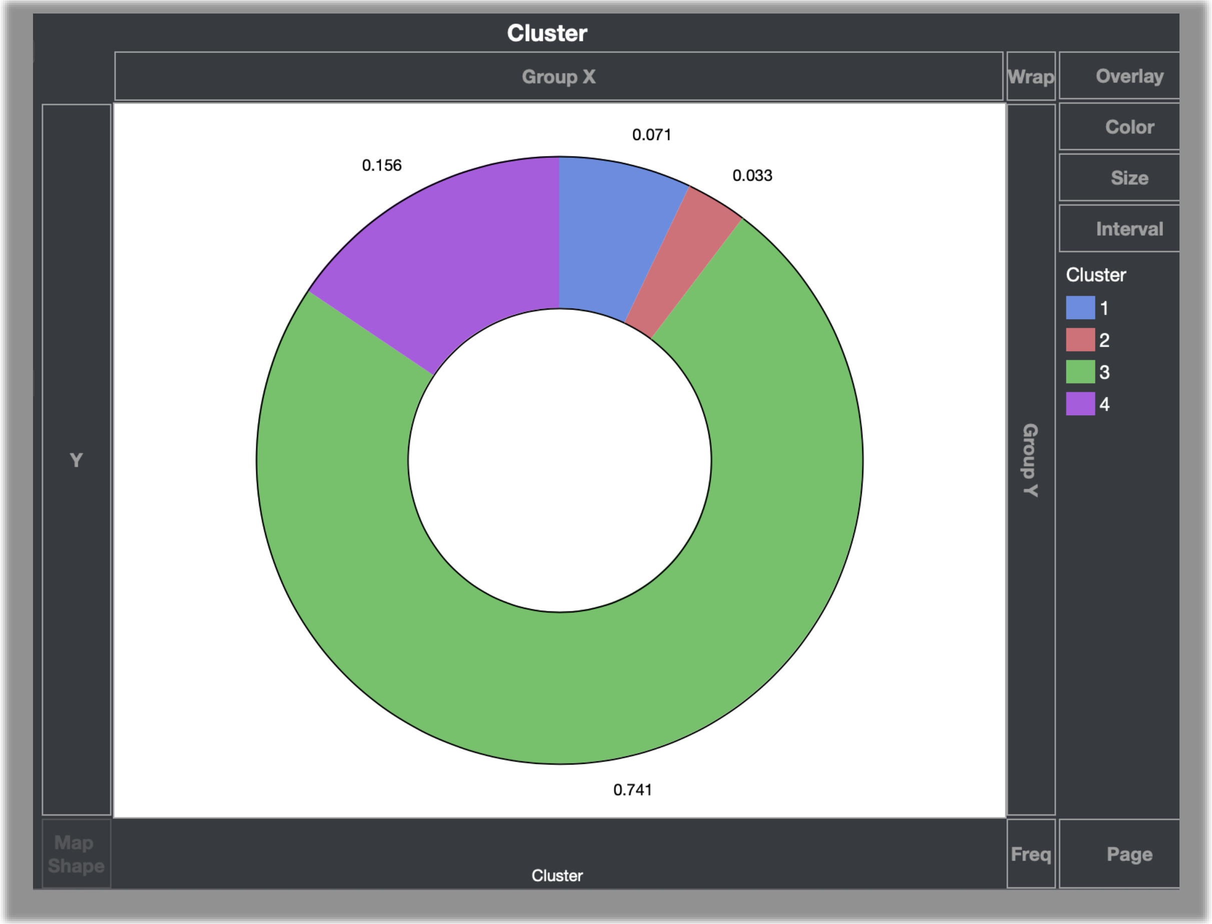
Cluster Development



- Cluster 4 (Premium Host Quality) are limited: only around 15.6% of listings
- Cluster 3 dominates the market at 74.1% of listings: which shows most listings are standard, smaller units

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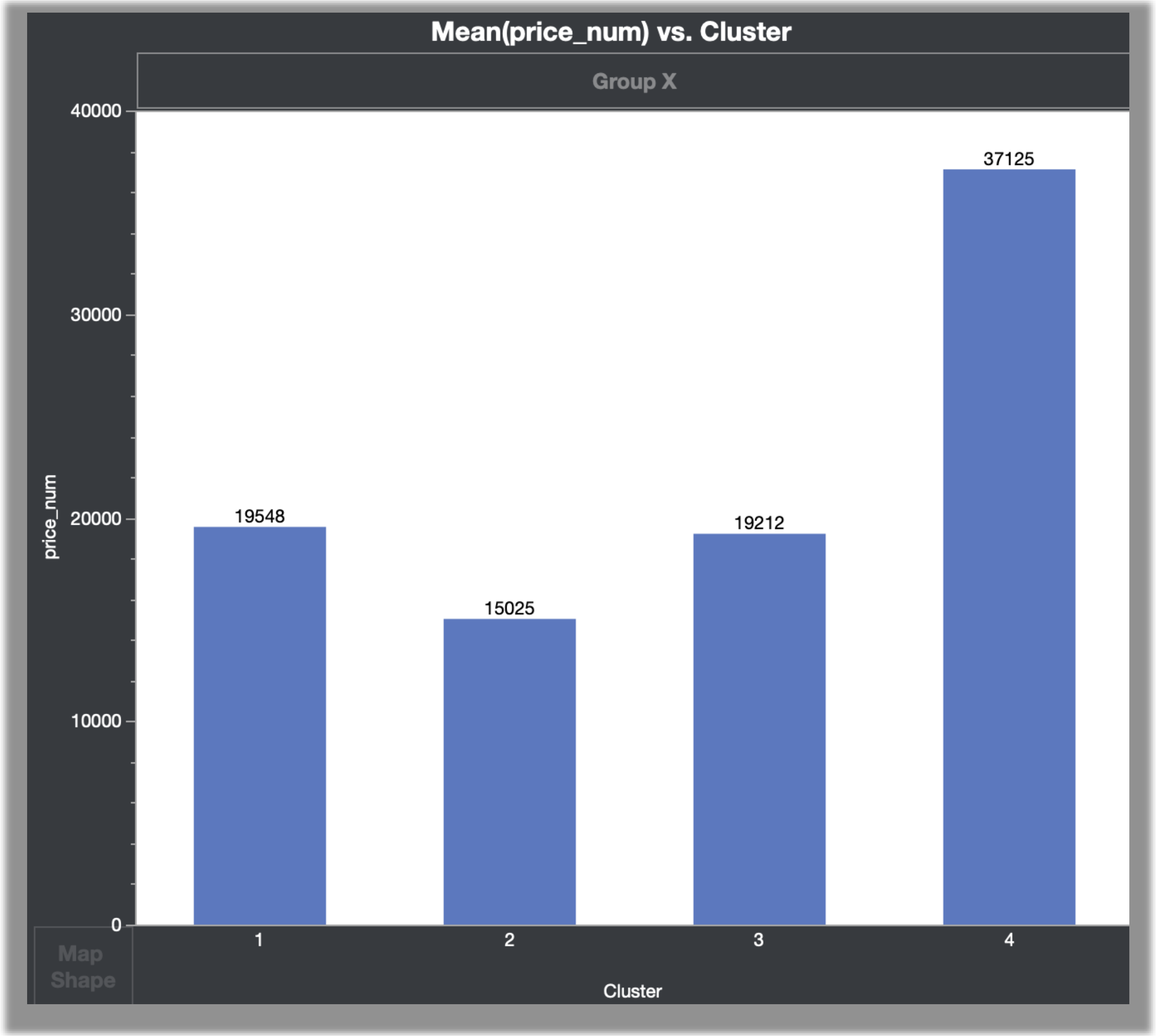
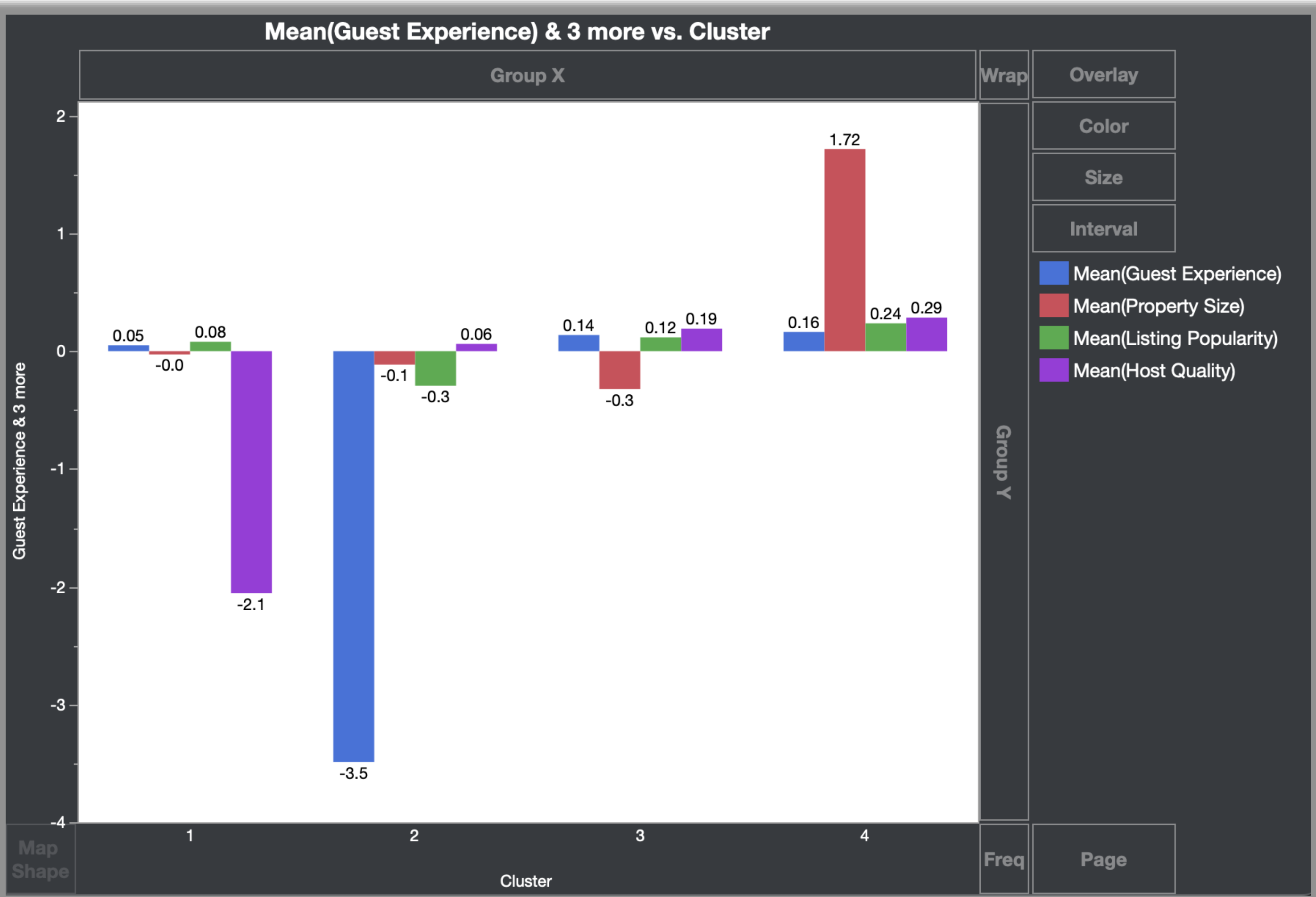
Cluster Vs. Price

- Cluster 4 is the most expensive of the group, but scored higher on property size
- Splitting the price among multiple travelers is an effective option





Cluster Development

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Valuable Cluster Insights

<u>Cluster 2</u>

- Far less guest satisfaction while still having a higher average price for a stay

<u>Cluster 3</u>

- Dominates the market at 74% of listings
- Most Tokyo Airbnb's are standard/smaller units

<u>Cluster 4</u>

- Strongest overall value
  - Has larger avg property size, positive guest experience, and stronger host quality
  - While sacrificing cheap price

Recommendations?

<u>Cluster 2</u>

- The least desired choice
- Only thing that could be considered is if it was discounted
- Average risk for bad service is higher

<u>Cluster 3</u>

- An amazing default option, 74% of market for a reason
- Often Budget friendly and gives standard expectations

<u>Cluster 4</u>

- Listing within this cluster are nice when budget allows
- You will often get a nice combination of quality and size

Classification Problem Setup

- Can we predict high-value stays using listing features like price, capacity, room type, ward, reviews, and availability?
- **Target variable:**
  High-value stay = 1 if the listing is both:

At or below the median nightly price

At or above the median review value score

- **Predictors used:**

Capacity, bedrooms, beds, bathrooms, room type, ward, availability, reviews, host response rate, host acceptance rate, and instant booking.

- **Models tested:**

Logistic Regression

Decision Tree

- **Evaluation metrics:**

AUC

Misclassification Rate

Accuracy

70/30 Training/Validation Split

Model 1: Logistic Regression Results

- Model has a overall classification accuracy of 78.88% (Misclassification Rate: 21.12%): high accuracy
- Geographic location (neighbourhood_cleansed) and property size (accommodates) are most significant drivers of value

Nominal Logistic Fit for High_Value_Stay Student Edition

Effect Summary

Source	Logworth	PValue
neighbourhood_cleansed	270.555	0.00000
accommodates	105.371	0.00000
room_type	28.777	0.00000
instant_bookable_dummy	24.472	0.00000
bedrooms_clean	15.546	0.00000
bathrooms_num	6.914	0.00000
availability_365	5.499	0.00000
host_acceptance_rate_num	3.134	0.00074
number_of_reviews	2.569	0.00270
host_response_rate_num	1.546	0.02842
beds_clean	0.542	0.28692

Remove Add Edit Exclude FDR

Converged in Gradient, 17 iterations

Iterations

Whole Model Test

Model	-LogLikelihood	DF	ChiSquare	Prob>ChiSq
Difference	1980.815	60	3961.629	<.0001*
Full	10342.365			
Reduced	12323.180			

RSquare (U)	0.1607
AICc	20807.1
BIC	21297.9
Observations (or Sum Wgts)	23189

Fit Details

Measure	Training	Definition
Entropy RSquare	0.1607	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.2399	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.4460	∑ -Log(ρ[j])/n
RASE	0.3812	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2902	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.2112	∑ (ρ[j]≠ρMax)/n
N	23189	n

Model 2: Decision Tree Results

Decision tree has a Misclassification Rate of: 21.19%

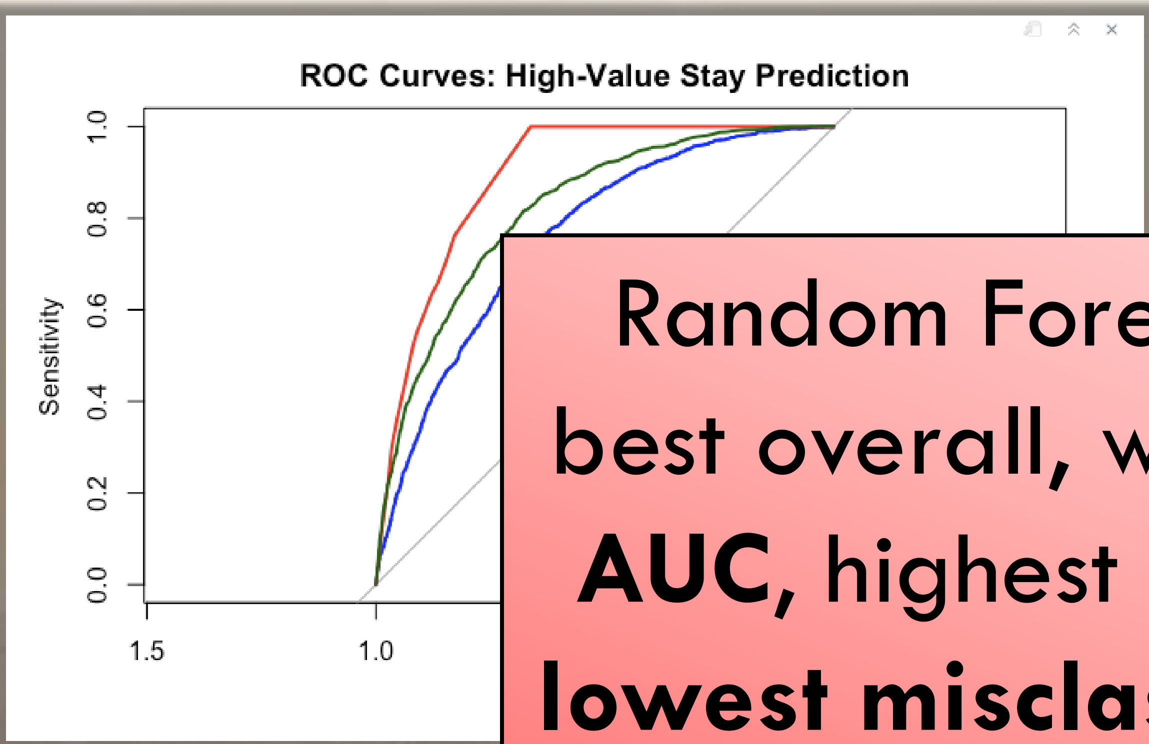
Overall Tree Accuracy: 78.81%

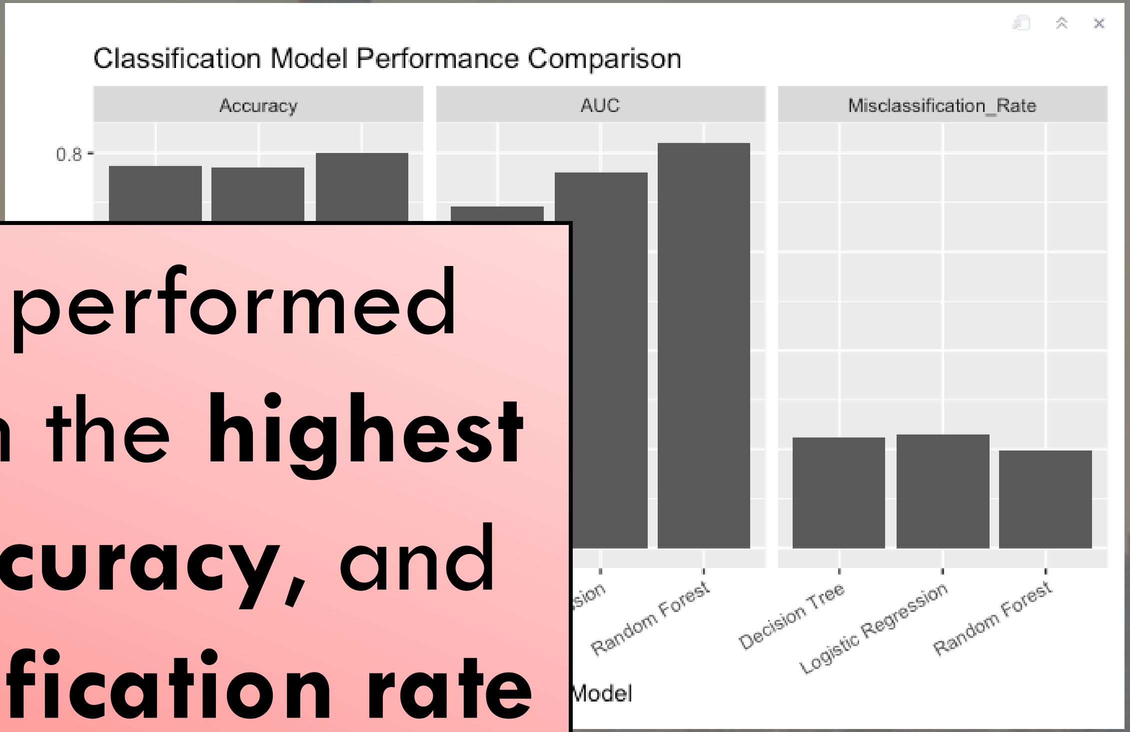
RSquare	N	Number of Splits
0.117	24721	4

Fit Details

Measure	Training	Definition
Entropy RSquare	0.1170	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.1768	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.4560	∑ -Log(ρ[j])/n
RASE	0.3847	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2960	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.2119	∑ (ρ[j]≠ρMax)/n
N	24721	n

Model Comparison in R





Random Forest performed best overall, with the **highest AUC**, highest **accuracy**, and **lowest misclassification rate**

Model	AUC	MCR	Accuracy
Random Forest	0.821	0.199	0.801
Logistic Regression	0.761	0.229	0.771
Decision Tree	0.693	0.225	0.775

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What do English guest reviews reveal about what travelers actually value?

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detected_langua... <chr>	Rows <int>	Percent <dbl>
English	595632	59.27
Blank / too short	141966	14.13
Japanese	114109	11.35
Chinese / CJK	98980	9.85
Korean	48716	4.85
Thai	2327	0.23
Latin / unknown	1843	0.18
Cyrillic / Russian-...	950	0.09
Other / unknown	362	0.04
Arabic	126	0.01

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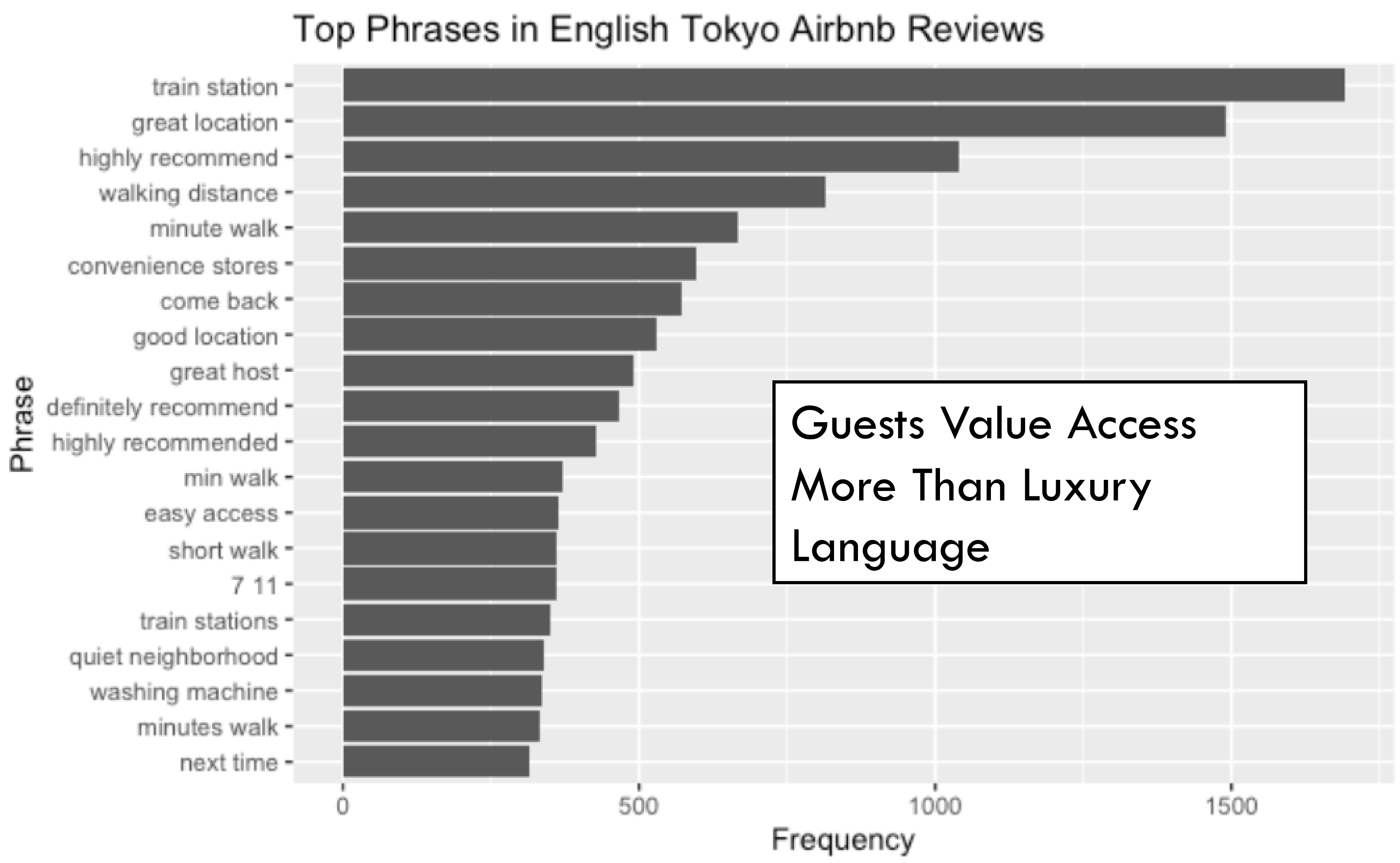

Guests Value Access More Than Luxury Language

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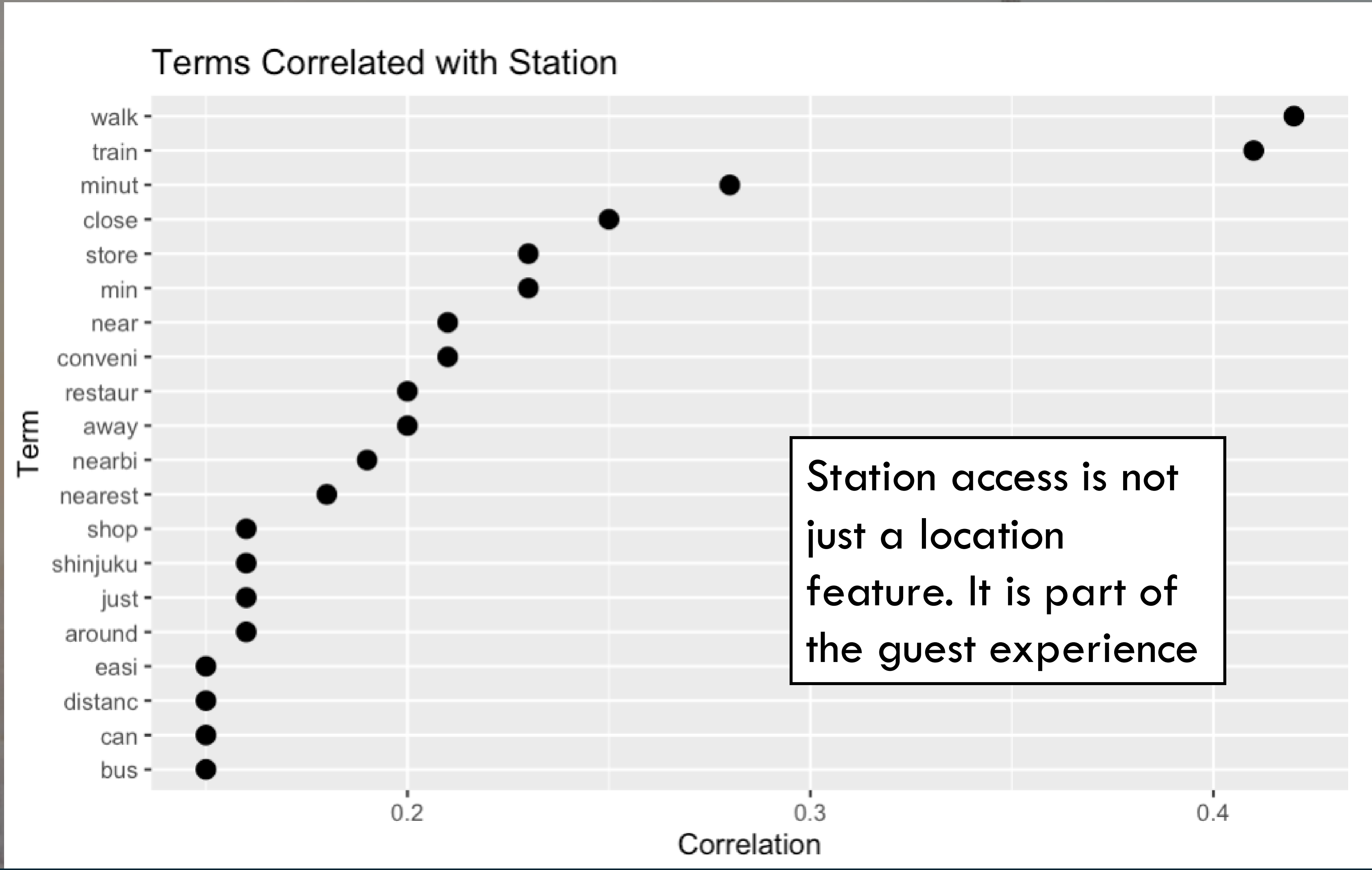

Station access is not just a location feature. It is part of the guest experience

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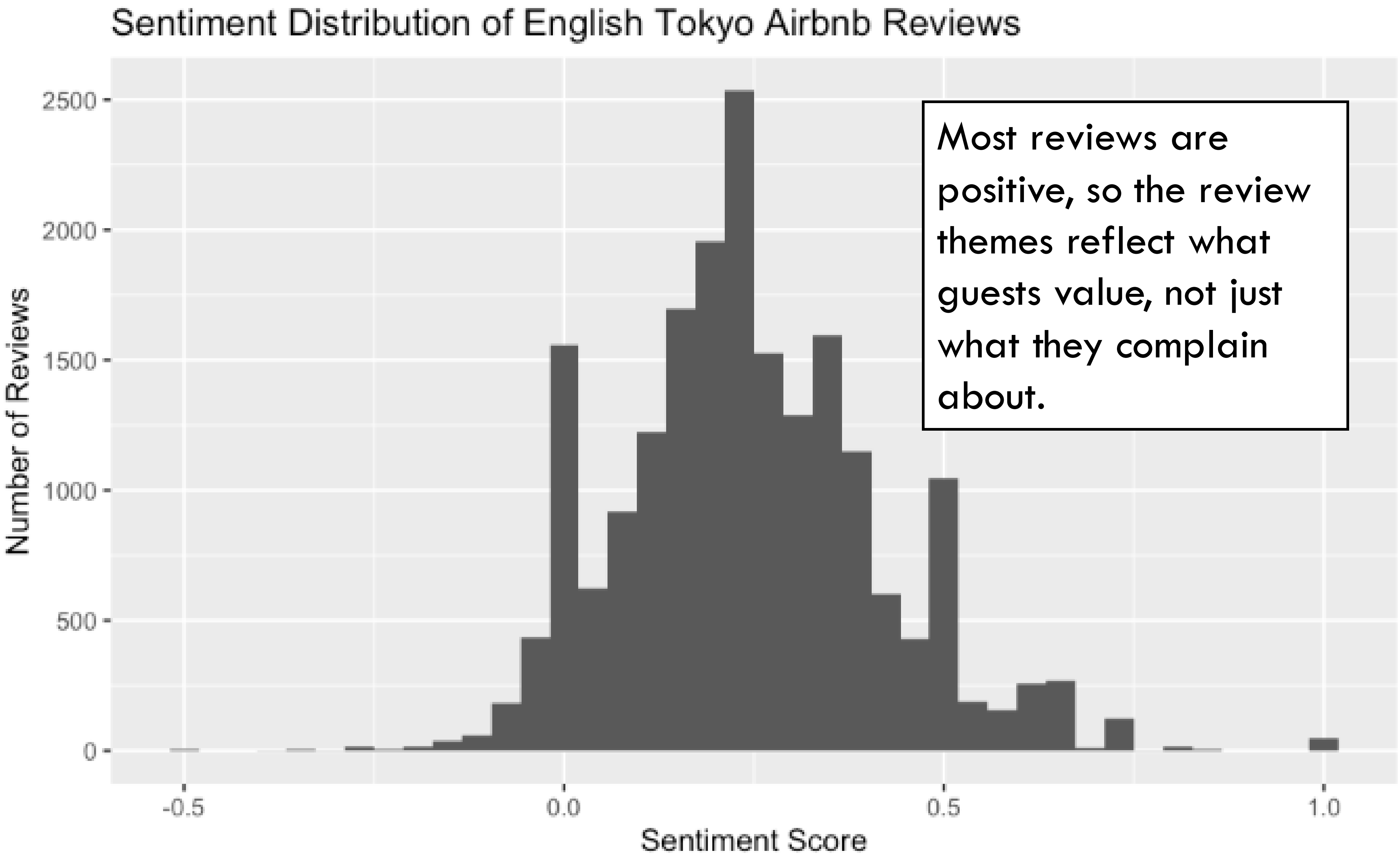
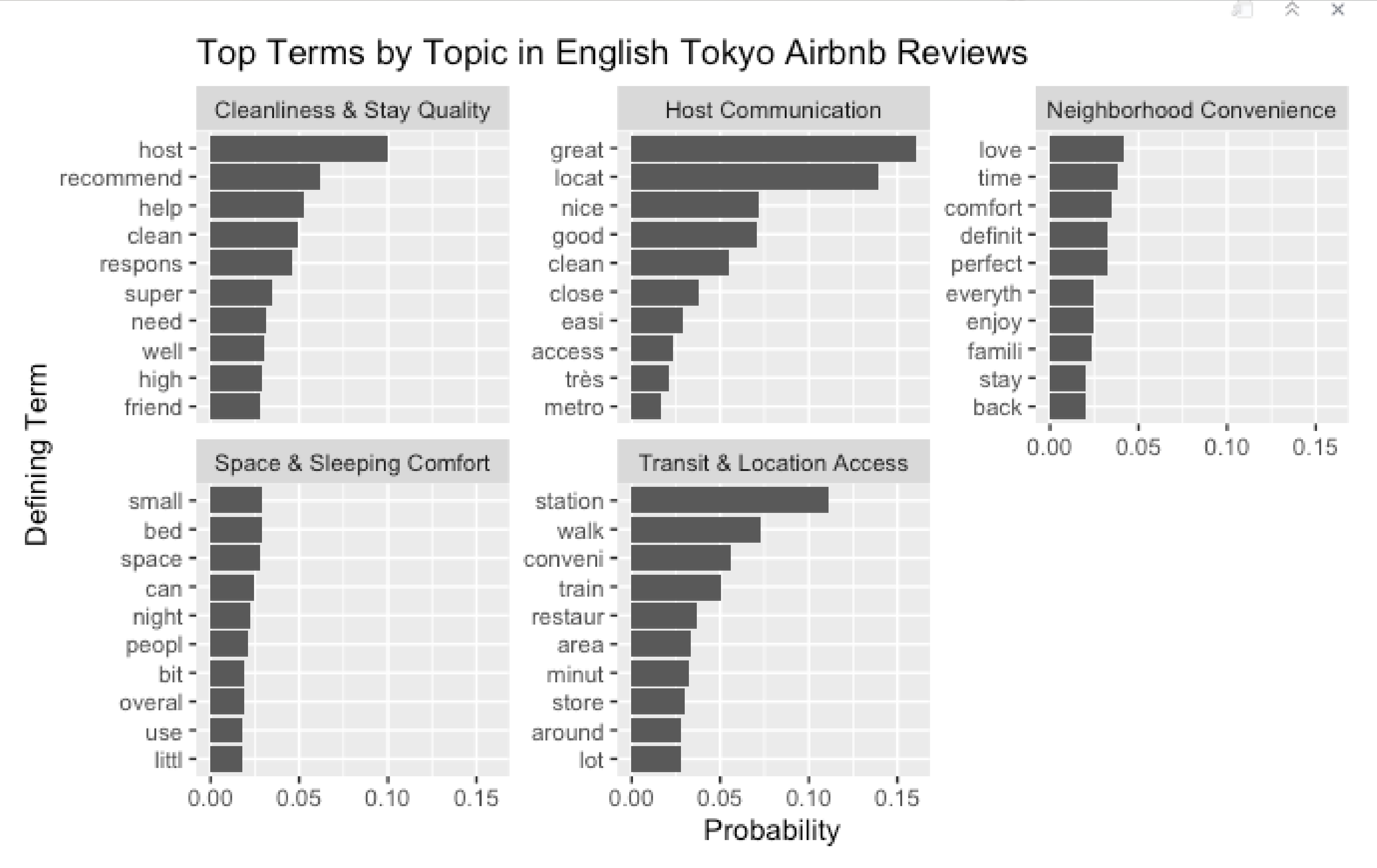


Most reviews are positive, so the review themes reflect what guests value, not just what they complain about.

LDA Topic Modeling Summary

Five recurring guest experience themes in English Tokyo Airbnb reviews

Topic	Marketing Label	Top Terms
1	Space & Sleeping Comfort	small, bed, space, can, night, peopl, bit, overal, use, littl
2	Transit & Location Access	station, walk, conveni, train, restaur, area, minut, store, around, lot
3	Cleanliness & Stay Quality	host, recommend, help, clean, respons, super, need, well, high, friend
4	Host Communication	great, locat, nice, good, clean, close, easi, access, très, metro
5	Neighborhood Convenience	love, time, comfort, definit, perfect, everyth, enjoy, famili, stay, back



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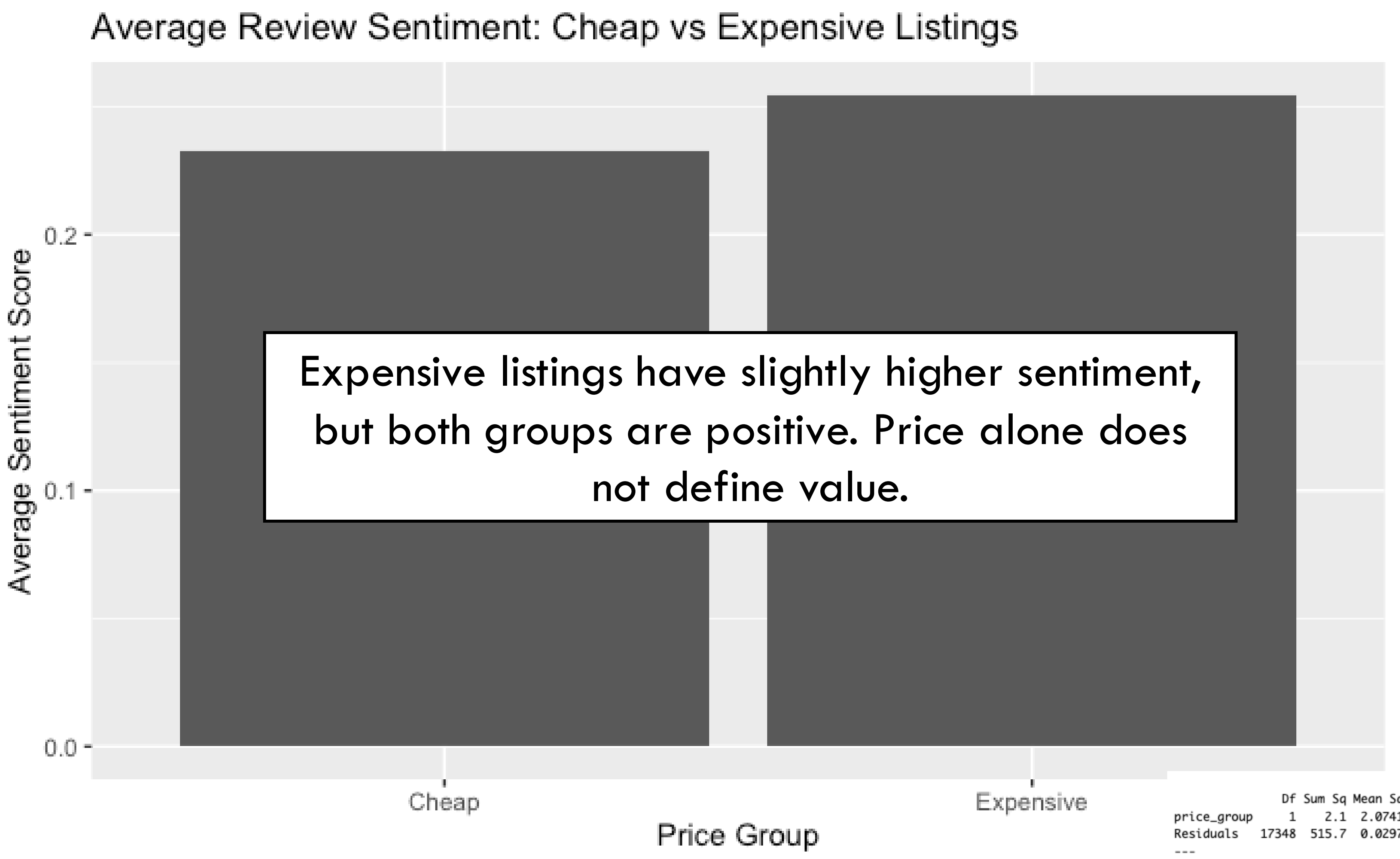

Expensive listings have slightly higher sentiment, but both groups are positive. Price alone does not define value.

```
             Df Sum Sq Mean Sq F value Pr(>F)
price_group      1    2.1  2.0741   69.78 <2e-16 ***
Residuals    17348  515.7  0.0297
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```
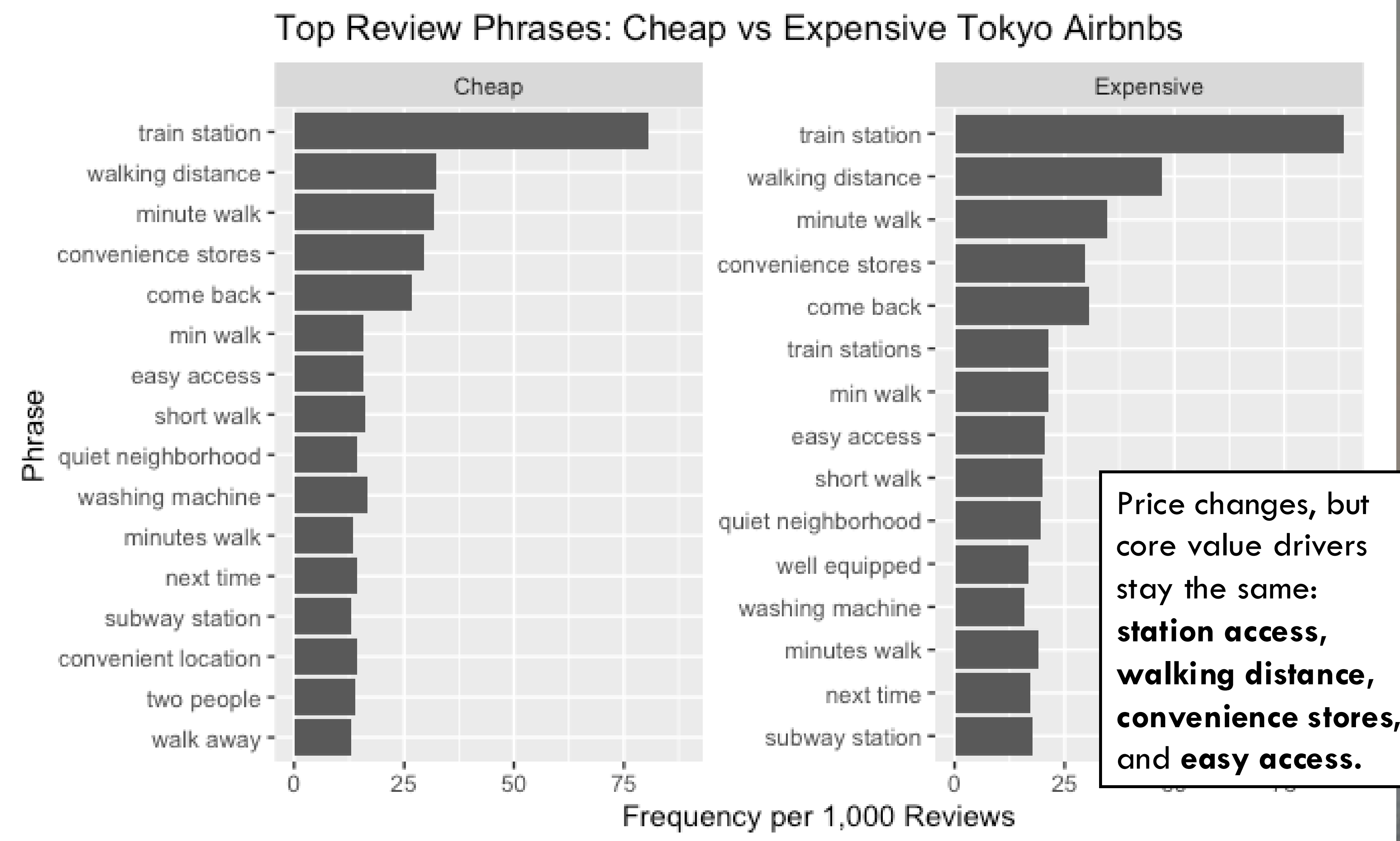
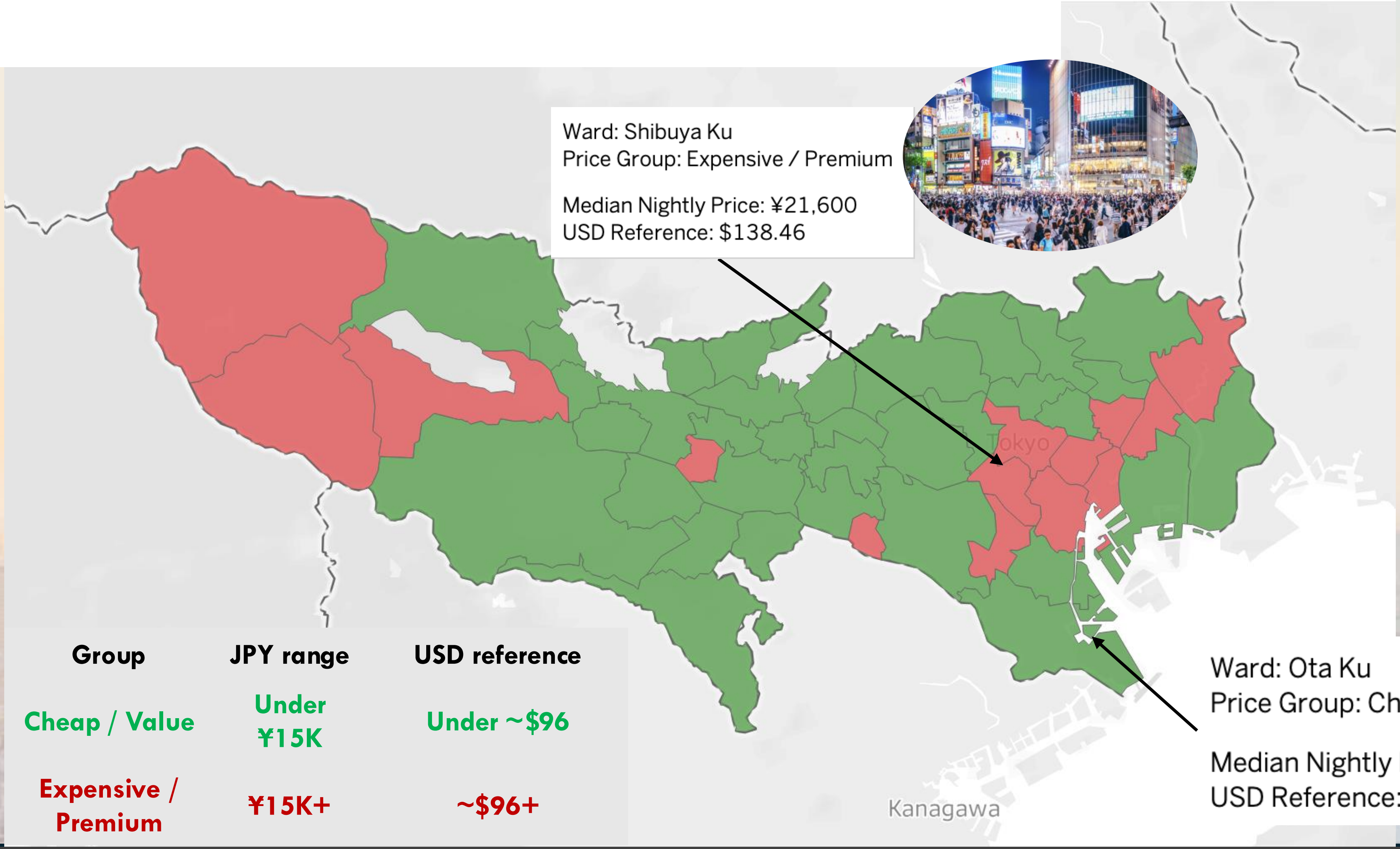



Price changes, but core value drivers stay the same: **station access, walking distance, convenience stores,** and **easy access.**

CHEAP VS EXPENSIVE MAP



Ward: Shibuya Ku
Price Group: Expensive / Premium

Median Nightly Price: ¥21,600
USD Reference: $138.46

Ward: Ota Ku
Price Group: Cheap / Value

Median Nightly Price: ¥10,800
USD Reference: $69.23

Group	JPY range	USD reference
Cheap / Value	Under ¥15K	Under ~$96
Expensive / Premium	¥15K+	~$96+

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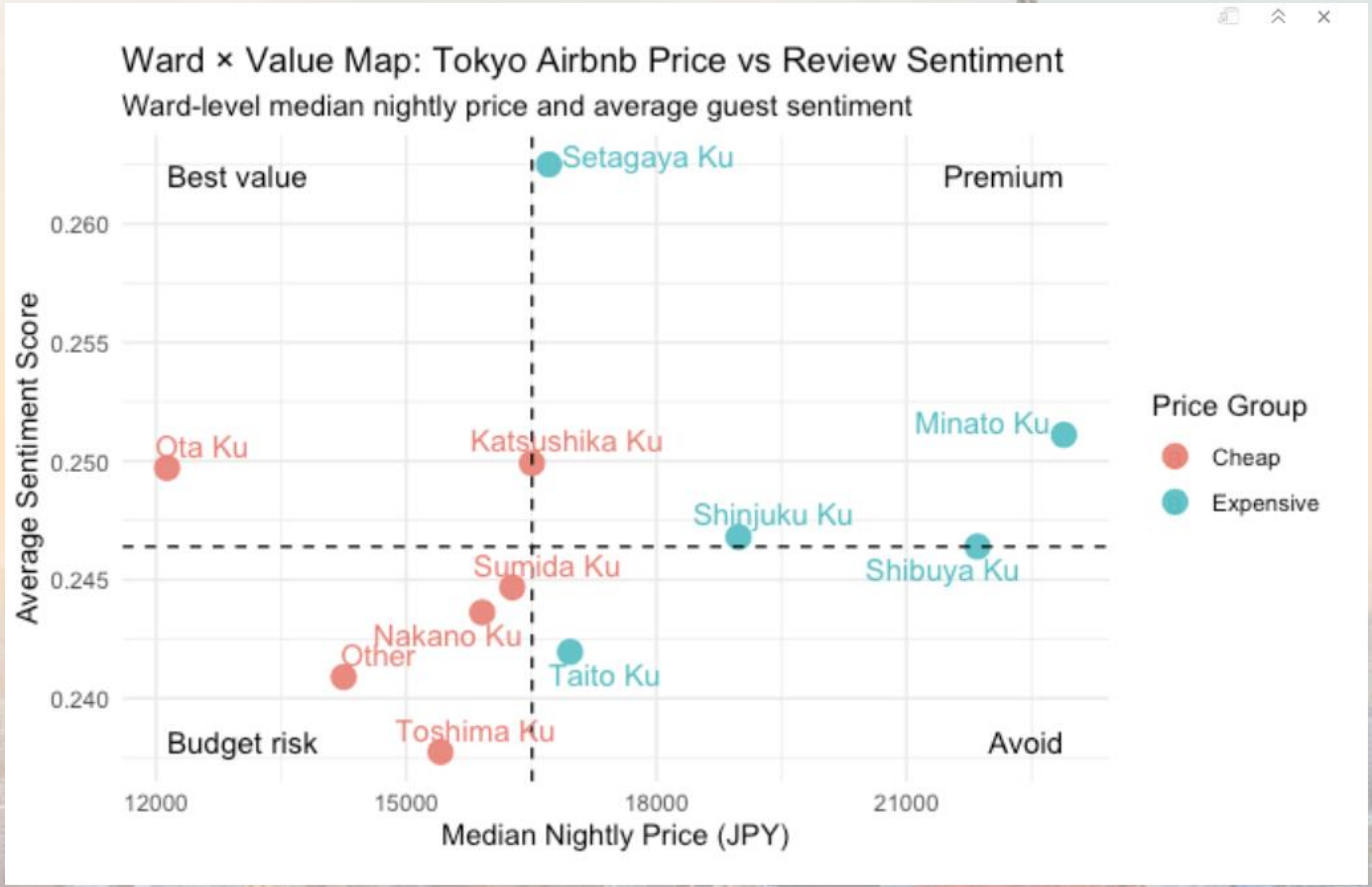

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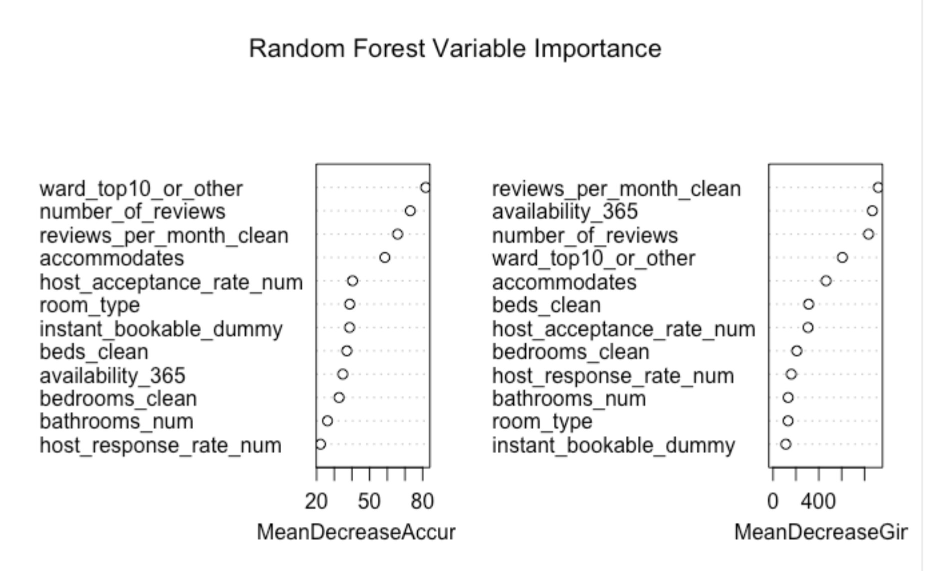
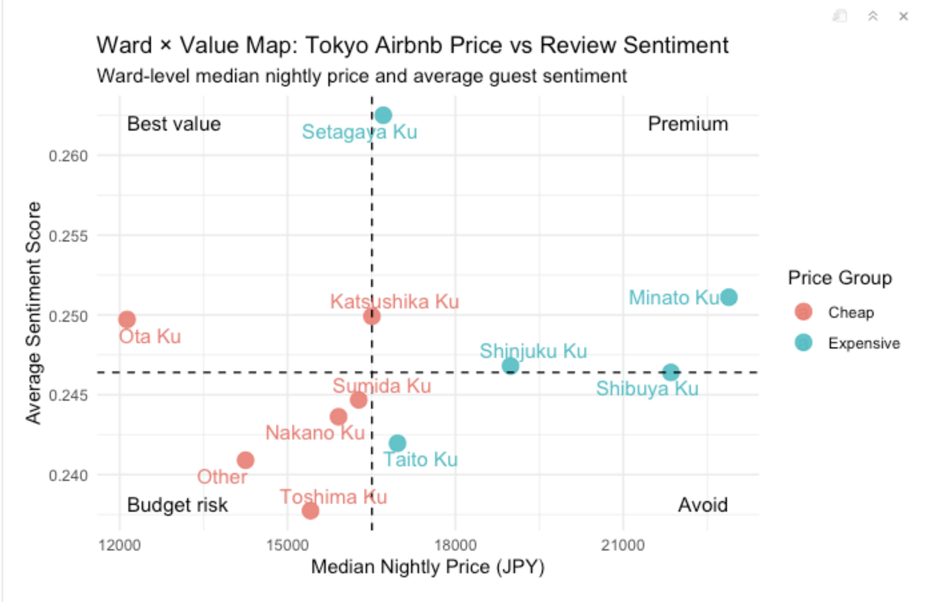
Final Recommendations & Conclusion

For Post-Grad Students…

- Ideal stay price/type → entire home under:
  ¥15,000 / ~$96
- Focus on cleanliness and good host reviews
- Convenience focused:
  - Near Train Stations
  - Close Airport Access
  - “Walking Distance”

Appendix/Extra Stuff





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